Statistics
Parameter Estimation, Hypothesis Testing, Regression

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Recap of Sample Mean and Sample Variance

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Sample Mean

- The expectation and variance of the sample mean are

$$\mathsf{E}(\hat{\mu}_n) = \mathsf{E}\left(\frac{\sum_{i=1}^{n} \mathsf{X}_i}{n}\right)$$

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Sample Mean

- The expectation and variance of the sample mean are

$$\mathsf{E}(\hat{\mu}_n) = \mathsf{E}\left(\frac{\sum_{i=1}^{n} \mathsf{X}_i}{n}\right) = \frac{\sum_{i=1}^{n} \mathsf{E}(\mathsf{X}_i)}{n}$$

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Sample Mean

- The expectation and variance of the sample mean are

$$\mathsf{E}(\hat{\mu}_n) = \mathsf{E}\left(\frac{\sum_{i=1}^{n} \mathsf{X}_i}{n}\right) = \frac{\sum_{i=1}^{n} \mathsf{E}(\mathsf{X}_i)}{n} = \mu$$

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Sample Mean

- The expectation and variance of the sample mean are

$$\begin{aligned} \mathsf{E}(\hat{\mu}_n) &= \mathsf{E}\left(\frac{\sum_{i=1}^{n} \mathsf{X}_i}{n}\right) = \frac{\sum_{i=1}^{n} \mathsf{E}(\mathsf{X}_i)}{n} = \mu \\ \mathsf{VAR}(\hat{\mu}_n) &= \mathsf{VAR}\left(\frac{\sum_{i=1}^{n} \mathsf{X}_i}{n}\right) \end{aligned}$$

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Sample Mean

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Sample Mean

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$$\begin{aligned} \mathsf{E}(\hat{\mu}_n) &= \mathsf{E}\left(\frac{\sum_{i=1}^n \mathsf{X}_i}{n}\right) = \frac{\sum_{i=1}^n \mathsf{E}(\mathsf{X}_i)}{n} = \mu \\ \mathsf{VAR}(\hat{\mu}_n) &= \mathsf{VAR}\left(\frac{\sum_{i=1}^n \mathsf{X}_i}{n}\right) = \frac{\sum_{i=1}^n \mathsf{VAR}(\mathsf{X}_i)}{n} = \frac{\sigma^2}{n} \end{aligned}$$

- What is the distribution of $\hat{\mu}_n$?

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Sample Mean

- The expectation and variance of the sample mean are

$$\begin{aligned} \mathsf{E}(\hat{\mu}_n) &= \mathsf{E}\left(\frac{\sum_{i=1}^n \mathsf{X}_i}{n}\right) = \frac{\sum_{i=1}^n \mathsf{E}(\mathsf{X}_i)}{n} = \mu \\ \mathsf{VAR}(\hat{\mu}_n) &= \mathsf{VAR}\left(\frac{\sum_{i=1}^n \mathsf{X}_i}{n}\right) = \frac{\sum_{i=1}^n \mathsf{VAR}(\mathsf{X}_i)}{n} = \frac{\sigma^2}{n} \end{aligned}$$

- What is the distribution of $\hat{\mu}_n$?
- How about 'an approximate distribution' of $\hat{\mu}_n$?

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Sample Variance: Known mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$\mathsf{V} \quad := \quad \sum_{i=1}^{n} (\mathsf{X}_i - \mu)^2$$

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Sample Variance: Known mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$\begin{aligned} V &:= \sum_{i=1}^{n} (X_i - \mu)^2 \\ E(V) &= E\left(\sum_{i=1}^{n} (X_i - \mu)^2 \right) \end{aligned}$$

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Sample Variance: Known mean

- Consider the sum of the square of the deviations from the mean, i.e.,

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\begin{aligned}
V &:= \sum_{i=1}^{n} (X_i - \mu)^2 \\
E(V) &= E\left(\sum_{i=1}^{n} (X_i - \mu)^2 \right) \\
&= \sum_{i=1}^{n} E\left((X_i - \mu)^2 \right)
\end{aligned}
$$

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Sample Variance: Known mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$\begin{aligned} \mathsf{V} &:= \sum_{i=1}^{n} (\mathsf{X}_i - \mu)^2 \\ \mathsf{E}(\mathsf{V}) &= \mathsf{E}\left(\sum_{i=1}^{n} (\mathsf{X}_i - \mu)^2 \right) \\ &= \sum_{i=1}^{n} \mathsf{E}\left((\mathsf{X}_i - \mu)^2 \right) \\ &= n\sigma^2 \end{aligned}$$

- Thus if μ is known then V/n is an unbiased estimator of $\mathsf{VAR}(\mathsf{X}_1)$.

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Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$V = \sum_{i=1}^{n} (X_i - \hat{\mu})^2$$

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Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$\begin{aligned} V &= \sum_{i=1}^{n} (X_i - \hat{\mu})^2 \\ &= \sum_{i=1}^{n} \left(X_i^2 - 2X_i\hat{\mu} + (\hat{\mu})^2 \right) \end{aligned}$$

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Sample Variance: Unknown mean

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&= \sum_{i=1}^{n} (X_i^2) - 2\hat{\mu} \sum_{i=1}^{n} X_i + \sum_{i=1}^{n} (\hat{\mu})^2
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&= \sum_{i=1}^{n} \left(X_i^2 \right) - 2\hat{\mu} \sum_{i=1}^{n} X_i + \sum_{i=1}^{n} (\hat{\mu})^2 \\
&= \sum_{i=1}^{n} \left(X_i^2 \right) - 2\hat{\mu}(n\hat{\mu}) + n\,(\hat{\mu})^2
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Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$\begin{aligned} V &= \sum_{i=1}^{n} (X_i - \hat{\mu})^2 \\ \sum_{i=1}^{n} (X_i - \hat{\mu})^2 &= \sum_{i=1}^{n} X_i^2 - n\,(\hat{\mu})^2 \end{aligned}$$

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Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$\begin{aligned} V &= \sum_{i=1}^{n} (X_i - \hat{\mu})^2 \\ \sum_{i=1}^{n} (X_i - \hat{\mu})^2 &= \sum_{i=1}^{n} X_i^2 - n(\hat{\mu})^2 \\ E(V) &= E\left(\sum_{i=1}^{n} X_i^2 - n(\hat{\mu})^2 \right) \end{aligned}$$

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E(V) &= E\left(\sum_{i=1}^{n} X_i^2 - n(\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} E(X_i^2) - nE((\hat{\mu})^2)
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E(V) &= E\left(\sum_{i=1}^{n} X_i^2 - n\,(\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} E(X_i^2) - nE((\hat{\mu})^2) \\
&= nVAR(X_1) + n(E(X_1))^2 - nVAR(\hat{\mu}) - n(E(\hat{\mu}))^2
\end{aligned}
$$

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E(V) &= E\left(\sum_{i=1}^{n} X_i^2 - n(\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} E(X_i^2) - nE((\hat{\mu})^2) \\
&= nVAR(X_1) + n(E(X_1))^2 - nVAR(\hat{\mu}) - n(E(\hat{\mu}))^2 \\
&= n\sigma^2 + n\mu^2 - n\left(\frac{\sigma^2}{n}\right) - n\mu^2
\end{aligned}
$$

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&= nVAR(X_1) + n(E(X_1))^2 - nVAR(\hat{\mu}) - n(E(\hat{\mu}))^2 \\
&= n\sigma^2 + n\mu^2 - n\left(\frac{\sigma^2}{n}\right) - n\mu^2 \; =
\end{aligned}
$$

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Sample Variance: Unknown mean

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\begin{aligned}
\mathsf{V} &= \sum_{i=1}^{n} (\mathsf{X}_i - \hat{\mu})^2 \\
\sum_{i=1}^{n} (\mathsf{X}_i - \hat{\mu})^2 &= \sum_{i=1}^{n} \mathsf{X}_i^2 - n\,(\hat{\mu})^2 \\
\mathsf{E}(\mathsf{V}) &= \mathsf{E}\left(\sum_{i=1}^{n} \mathsf{X}_i^2 - n\,(\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} \mathsf{E}(\mathsf{X}_i^2) - n\mathsf{E}((\hat{\mu})^2) \\
&= n\mathsf{VAR}(\mathsf{X}_1) + n(\mathsf{E}(\mathsf{X}_1))^2 - n\mathsf{VAR}(\hat{\mu}) - n(\mathsf{E}(\hat{\mu}))^2 \\
&= n\sigma^2 + n\mu^2 - n\left(\frac{\sigma^2}{n}\right) - n\mu^2 \;=\; (n-1)\sigma^2.
\end{aligned}
$$

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Sample Variance: Unknown mean

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\begin{aligned}
\mathsf{V} &= \sum_{i=1}^{n} (\mathsf{X}_i - \hat{\mu})^2 \\
\sum_{i=1}^{n} (\mathsf{X}_i - \hat{\mu})^2 &= \sum_{i=1}^{n} \mathsf{X}_i^2 - n\,(\hat{\mu})^2 \\
\mathsf{E}(\mathsf{V}) &= \mathsf{E}\left(\sum_{i=1}^{n} \mathsf{X}_i^2 - n\,(\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} \mathsf{E}(\mathsf{X}_i^2) - n\mathsf{E}((\hat{\mu})^2) \\
&= n\mathsf{VAR}(\mathsf{X}_1) + n(\mathsf{E}(\mathsf{X}_1))^2 - n\mathsf{VAR}(\hat{\mu}) - n(\mathsf{E}(\hat{\mu}))^2 \\
&= n\sigma^2 + n\mu^2 - n\left(\frac{\sigma^2}{n}\right) - n\mu^2 \;=\; (n-1)\sigma^2.
\end{aligned}
$$

- Thus, when μ is not known, $\frac{V}{n-1}$ is the unbiased estimate of $\mathsf{VAR}(\mathsf{X}_1)$.

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Lecture 3

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Summarizing...

- Random independent samples from a population

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Summarizing...

- Random independent samples from a population
- Each sample is an independent random variable

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Summarizing...

- Random independent samples from a population
- Each sample is an independent random variable
- Sample mean is a statistic.

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Summarizing...

- Random independent samples from a population
- Each sample is an independent random variable
- Sample mean is a statistic.
- Expectation of sample mean is the same as the true mean.

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Summarizing...

- Random independent samples from a population
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- Sample mean is a statistic.
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- Variance of the sample mean is σ^2/n

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Summarizing...

- Random independent samples from a population
- Each sample is an independent random variable
- Sample mean is a statistic.
- Expectation of sample mean is the same as the true mean.
- Variance of the sample mean is σ^2/n
- Since samples are independent, we can invoke LLN and CLT on the Sample Sum.

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CLT and the Sample Mean

- Let μ and σ^2 be the true mean and true variance of a population feature.

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CLT and the Sample Mean

- Let μ and σ^2 be the true mean and true variance of a population feature.
- Applying CLT to the sample sum of the n independent sample, S_n we have

$$\frac{S_n - n\mu}{\sqrt{n}\sigma} \rightarrow \mathsf{Z}$$

where Z is the unit normal.

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$$\frac{S_n - n\mu}{\sqrt{n}\sigma} \rightarrow \mathsf{Z}$$

  where Z is the unit normal.
- Let $\mathsf{Prob}(\mathsf{Z} > z_\alpha) = \alpha$

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CLT and the Sample Mean

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- Let $\mathsf{Prob}(\mathsf{Z} > z_\alpha) = \alpha$

$$\frac{S_n - n\mu}{\sqrt{n}\sigma} = n\frac{S_n/n - \mu}{\sqrt{n}\sigma}$$

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- Let $\mathsf{Prob}(\mathsf{Z} > z_\alpha) = \alpha$

$$\frac{S_n - n\mu}{\sqrt{n}\sigma} = n\frac{S_n/n - \mu}{\sqrt{n}\sigma} = \frac{\hat{\mu}_n - \mu}{(\sigma/\sqrt{n})}$$

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CLT and the Sample Mean

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$$\frac{S_n - n\mu}{\sqrt{n}\sigma} = n\frac{S_n/n - \mu}{\sqrt{n}\sigma} = \frac{\hat{\mu}_n - \mu}{(\sigma/\sqrt{n})}$$

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CLT and the Sample Mean

- Let us now characterize deviation of sample mean from true mean.

$$\text{Prob}\left(-z_\alpha \leq \frac{\hat{\mu}_n - \mu}{(\sigma/\sqrt{n})} \leq z_\alpha\right) = 1 - 2\alpha$$

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$$\text{Prob}\left(-\frac{\sigma}{\sqrt{n}} z_\alpha \leq (\hat{\mu}_n - \mu) \leq z_\alpha \frac{\sigma}{\sqrt{n}}\right) = 1 - 2\alpha$$

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CLT and the Sample Mean

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$$\text{Prob}\left(-z_\alpha \leq \frac{\hat{\mu}_n - \mu}{(\sigma/\sqrt{n})} \leq z_\alpha\right) = 1 - 2\alpha$$

$$\text{Prob}\left(-\frac{\sigma}{\sqrt{n}} z_\alpha \leq (\hat{\mu}_n - \mu) \leq z_\alpha \frac{\sigma}{\sqrt{n}}\right) = 1 - 2\alpha$$

$$\text{Prob}\left(|\hat{\mu}_n - \mu| \leq z_\alpha \frac{\sigma}{\sqrt{n}}\right) = 1 - 2\alpha$$

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CLT and the Sample Mean

- Let us now characterize deviation of sample mean from true mean.

$$\begin{aligned}
\text{Prob}\left(-z_\alpha \leq \frac{\hat{\mu}_n - \mu}{(\sigma/\sqrt{n})} \leq z_\alpha\right) &= 1-2\alpha \\
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\text{Prob}\left(|\hat{\mu}_n - \mu| \leq z_\alpha \frac{\sigma}{\sqrt{n}}\right) &= 1-2\alpha \\
\text{Prob}\left(\hat{\mu}_n - \frac{\sigma}{\sqrt{n}} z_\alpha \leq \mu \leq \hat{\mu}_m + z_\alpha \frac{\sigma}{\sqrt{n}}\right) &= 1-2\alpha
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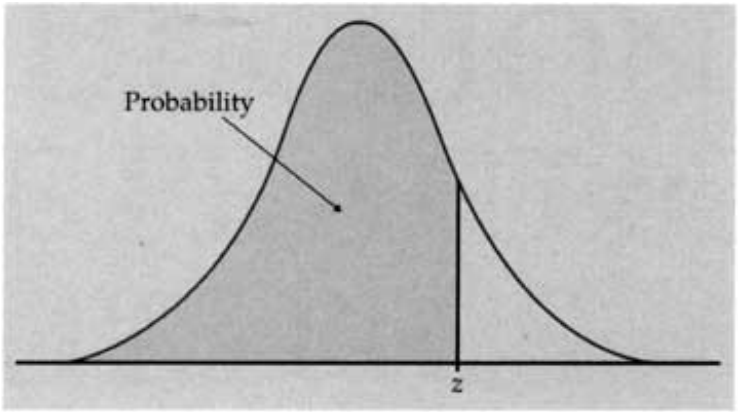


Table entry for z is the area under the standard normal curve to the left of z.

TABLE A Standard normal probabilities (*continued*)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986

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CLT and the Sample Mean

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- $\left(\hat{\mu}_n - \frac{\sigma}{\sqrt{n}}\, z_\alpha, \hat{\mu}_m + z_\alpha \frac{\sigma}{\sqrt{n}}\right)$ is called the $(1-\alpha)$ confidence interval for μ.

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- Usually, variance is not known! So ...
- Compute

$$\hat{V} = \frac{1}{n-1} \sum_{i=1}^{n} (\mathsf{X}_i - \hat{\mu})^2$$

and use $\hat{V}$ as the variance, i.e., use $\sigma = \sqrt{\hat{V}}$ in the preceding.

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The $n\sigma$ rules

- 3σ and 6σ rule is something that is commonly used in quality control departments.

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- Lower tolerance requires correspondingly lower σ and a tighter control over the process.

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Some experiments

- Write a program to do the following:

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- Write a program to do the following:
  1. Generate n IID random variables of a specified distribution F whose mean and standard deviation are known.

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  5. Plot the histogram of the sample mean from the K experiments.
  6. Make a scatter plot of the left and the right limits of the 95% confidence interval.

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Some experiments

- Let F correspond to Gaussian with $\mu = 0$ and $\sigma = 2$. Let $n = 20$ and $K = 1000$. Repeat for different $\sigma = 3, 5$ and $n = 50, 100, 200, 500, 1000$.

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- And for F being a Bernouilli distribution
- Now pretend σ is not known and use $\hat{V}$ as above and see if there is a difference.

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Sampling from a Normal Population

- If X_i are i.i.d. Gaussian with mean μ and standard deviation σ then

$$\frac{\hat{\mu} - \mu}{\sigma / \sqrt{n}}$$

is standard normal.

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- Define

$$S^2 = \frac{\sum_{i=1}^{n}(X_i - \hat{\mu})^2}{n-1}$$

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- Can show that $\hat{\mu}$ and S^2 are independent random variables.

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Known Variance

- Since $\hat{\mu}_n$ is Gaussian, we can say

$$\text{Prob}\left(\mu - z_{\delta/2}\frac{\sigma}{\sqrt{n}} \leq \hat{\mu}_n \leq \mu + z_{\delta/2}\frac{\sigma}{\sqrt{n}}\right) = 1 - \frac{\delta}{2} - \frac{\delta}{2}$$

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- In this case we say that $(\hat{\mu}_n - z_{\delta/2}\frac{\sigma}{\sqrt{n}})$ is the γ confidence interval for μ.

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- If the population is not from a normal distribution, invoke CLT to use the preceding as an approximation.

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- If the population is not from a normal distribution, invoke CLT to use the preceding as an approximation.
- Alternatively, use Chebyshev inequality to

$$\text{Prob}\left(\hat{\mu}_n - \frac{\sigma}{\sqrt{n\delta}} \leq \mu \leq \hat{\mu}_n + \frac{\sigma}{\sqrt{n\delta}}\right) \geq 1 - \delta = \gamma.$$

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Unknown Variance: Distributions of $\hat{\mu}$ and S^2

- X_i, $i = 1, \ldots, n$ are iid samples of Gaussian of mean μ and variance σ^2.

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- Sum of independent Gaussian random variables is also a Gaussian.
- Thus the sample mean $\hat{\mu}$, is Gaussian with mean μ and variance σ^2/n.
- Now consider

$$\begin{aligned} \frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2 &= \frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \hat{\mu})^2 + \frac{1}{\sigma^2} n(\hat{\mu} - \mu)^2 \\ &= \frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \hat{\mu})^2 + \left(\frac{\sqrt{n}(\hat{\mu} - \mu)}{\sigma}\right)^2 \end{aligned}$$

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Chi-Square and t Distributions

- If $Z, Z_1, \ldots Z_n$ are independent unit normal random variables then

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  - $Z_1^2 + Z_2^2 + \cdots + Z_n^2$ is a χ^2 with n-degrees of freedom. It has mean 0 and variance $2n$.

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  - The quantity

$$\frac{Z}{\sqrt{(Z_1^2 + Z_2^2 + \cdots + Z_n^2)/n}}$$

    has t-distribution. This has mean 0 and variance $n/(n-2)$.

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  - $Z_1^2 + Z_2^2 + \cdots + Z_n^2$ is a χ^2 with n-degrees of freedom. It has mean 0 and variance $2n$.
  - The quantity

$$\frac{Z}{\sqrt{(Z_1^2 + Z_2^2 + \cdots + Z_n^2)/n}}$$

    has t-distribution. This has mean 0 and variance $n/(n-2)$.
- LHS is χ_n^2. Second term of RHS is χ_1^2. Hence reasonable to guess that first term of RHS is χ_{n-1}^2.

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Unknown Variance: Distributions of $\hat{\mu}$ and S^2

- Can show that $\hat{\mu}$ and S^2 are independent with $\hat{\mu}$ being normal and $(n-1)S^2/\sigma^2$ being χ^2_{n-1}.

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- This helps in obtaining the confidence interval when the variance is unknown.

$$\text{Prob}\left(-t_{\alpha/2,n-1} < \sqrt{n}\frac{\hat{\mu}-\mu}{S} < t_{\alpha/2,n-1}\right) = 1-\alpha$$

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- To obtain confidence interval for vairance note that $(n-1)S^2/\sigma^2$ is χ^2_{n-1}. Thus

$$\text{Prob}\left(\chi^2_{\alpha/2,n-1} \leq (n-1)\frac{S^2}{\sigma^2} \leq \chi^2_{\alpha/2,n-1}\right) = 1-\alpha$$

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Parameter Estimation

- Many times, we have a model for the CDF of pdf of population feature of interest.

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- As before we will obtain n independent sample realizations of the parameter of interest.

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What is being estimated?



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What is being estimated?



And what is the method?

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Likelihood Function

- $x_1, \ldots x_K$ is the set of measurements.

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Likelihood Function

- $x_1, \ldots x_K$ is the set of measurements.
- Assume that the parameter set is θ.

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  - Assume the population is exponential with parameter μ.

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Maximum Likelihood Estimate

- Find the θ that maximises the likelihood of the observations.

$$\hat{\theta} = \arg\max_{\theta} L(x_1, \ldots, x_K|\theta)$$

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$$\begin{aligned} L(\underline{x}; p) &= \prod_{i=1}^{K} p^{x_i}(1-p)^{1-x_i} = p^n(1-p)^{K-n} \\ \hat{p} &= \frac{\sum_{i=1}^{K} x_i}{K} \end{aligned}$$

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